Springer–Borel maps

Integral images and the geometry of representation maps

A structural research program for Sean Rogers

Proposal and progress inventory, 20 October 2026

Purpose and central question

The published Springer-morphism and Levi-character papers make a representation-dependent map from characters to flag cohomology explicit. The proposed program asks for structural control of the arithmetic and geometry of that map:

> *How much of a Springer–Borel map can be predicted from the root datum, the representation, the character lattice and the Levi subgroup, before expanding its Schubert coefficients?*

The first objective is a theory of integral images. The second studies the generic degree and regularity of the underlying Springer morphism. These share the same explicit map but retain different information: the Borel quotient kills positive-degree global invariants, while the adjoint-quotient map preserves the invariant functions needed for degree calculations.

Aim	Goal	First meaningful deliverable
A	Explain primes, saturation and Smith factors in integral images.	A precise local comparison theorem and a source/parabolic criterion for possible torsion primes.
B	Compute generic degrees by a uniform, effective boundary method.	A certified face-scheme procedure, proved for a coherent family.
C	Determine when Springer maps preserve regular elements.	A reusable obstruction or sufficient criterion, beyond individual examples.
D	Give geometric meaning to Smith filtrations.	An integral comparison for one chain/incidence model, including primitivity.
E	Extend image questions to other theories and groups.	An equivariant pilot, followed by filtered K-theory and compatible quantum/affine constructions.

What is already available. The repository contains internally audited source-lattice arguments, general normalized type B and D single-tail Smith formulas, two-step joint-image theorems, exceptional integral-image calculations, and explicit quotient maps, critical determinants and degree calculations for selected exceptional representations. These are substantial starting points, with their precise scope recorded below. They are not all published results.

What success means. A satisfactory outcome is a theorem with checkable hypotheses, a proof of the mechanism, and independent exact examples. A counterexample delimiting a plausible generalization, or a theorem for a well-chosen family, also counts as progress. More tables without a question they distinguish are secondary. This proposal organizes research; it does not commit to writing a full paper or claim priority for the repository deductions. Track E is now part of the program, with six dependent subprojects rather than six simultaneous implementations.

1 Published foundations and common notation

Kumar introduced the map from polynomial characters of a Levi subgroup to flag cohomology [1]. Rogers determined the torus coordinates of the Springer morphism by its weights [2]. Kumar–Rogers, Part II, treat the symplectic and odd orthogonal constructions and stable injectivity; Kumar–Xie, Part III, supply the even orthogonal analogue and exceptional generator calculations [3, 4]. The present program refines the integral arithmetic and studies the geometry of the same trace map.

Let G initially be a simply connected simple complex group, with $T \subset B \subset G$, Weyl group W, and Lie algebra $\mathfrak{g}$. Write $X = X(T)$ for the weight lattice, Q_{rt} for the root lattice, and $V = X \otimes \mathbb{Q}$. Put $r = \operatorname{rank} G$. We reserve P for a standard parabolic, with Levi $L \supset T$ and Weyl group W_L. This avoids confusing the parabolic with a lattice. Other isogeny forms are separate extensions; the orthogonal papers use their specified SO forms.

For a representation ρ with faithful differential, $\theta_\rho : G \to \mathfrak{g}$ is normalized trace projection, so $d\theta_\rho|_1 = \mathrm{id}$. For irreducible $V(\lambda)$ write θ_λ. In the fundamental-weight/simple-coroot coordinates of Rogers's formula,

$$M_\rho = \sum_\mu m_\mu \mu\mu^t, \qquad \theta_\rho(t) = M_\rho^{-1} \sum_\mu m_\mu \mu e^\mu(t).$$

Here μ is a weight column, m_μ its multiplicity, and e^μ a Laurent character. This is a published coordinate method, not a new construction of this proposal.

The pullback embeds $\mathrm{Sym}_{\mathbb{C}}(\mathfrak{t}^*)^{W_L}$ into the character algebra. Its image is $\mathrm{Rep}^{\mathbb{C}}_{\lambda\text{-poly}}(L)$ in the papers. Their map is

$$\xi^P_\lambda = \beta^P \circ (\theta^*_\lambda)^{-1} : \mathrm{Rep}^{\mathbb{C}}_{\lambda\text{-poly}}(L) \longrightarrow H^*(G/P, \mathbb{C}).$$

The inverse is defined on the image of the injective pullback. It is not a birational inverse of θ_λ. The rational Borel map in polynomial coordinates is $\beta^P : \mathrm{Sym}_{\mathbb{Q}}(V)^{W_L} \to H^{2*}(G/P, \mathbb{Q})$. Define

$$\mathcal{C}_\rho = \{f \in \mathrm{Sym}_{\mathbb{Q}}(V) : \theta^*_\rho f \in \mathbb{Z}[X]\}, \quad \Lambda_\rho = (\mathcal{C}_\rho)_1, \quad \mathcal{A}_{\rho,L} = \mathcal{C}_\rho^{W_L}.$$

Thus the literal integral character source corresponds to $\mathcal{A}_{\rho,L}$. Where integrality of β^P on this source has been justified, put $J_{\rho,L} = \beta^P(\mathcal{A}_{\rho,L})$.

For any specified integral image $J_d \subset H^{2d}(G/P, \mathbb{Z})$, set

$$S_d = \mathrm{Sat}(J_d) = (J_d \otimes \mathbb{Q}) \cap H^{2d}(G/P, \mathbb{Z}), \qquad D_d = S_d/J_d.$$

The nonzero Smith factors describe D_d, including unit factors when recording ranks. For selected generator subalgebras, S_d need not be all cohomology. Polynomial degree d means cohomological degree $2d$; it is not Laurent total degree.

2 Completed source work: the starting point for Aim A

Status: The following are written, internally audited repository deductions. Their exact literature priority and external expert review remain unresolved.

2.1 The full torus source is controlled in degree one

Under the hypotheses above, including reducible ρ, the current argument gives

$$\boxed{\mathcal{C}_\rho = \mathrm{Sym}_{\mathbb{Z}}(\Lambda_\rho), \qquad \mathcal{A}_{\rho,L} = \mathrm{Sym}_{\mathbb{Z}}(\Lambda_\rho)^{W_L}.}$$

This includes inhomogeneous polynomials. The proof uses the full linear integrality lattice, primitive Laurent coefficient columns and the trace factorization to control independence modulo primes; it does not assume that the trace form remains nondegenerate in every characteristic. The theorem concerns characteristic-zero maps and integral coefficients, not a universal finite-field trace-projection construction. See the source theorem and its proof/source audit (ledger SRC-01/03).

This resolves the torus-source generation question. It does not imply that Levi invariants are generated by invariant linear elements, or that their Borel images are saturated. Those are the next stages of the program.

2.2 An irreducible highest-weight formula in every simple type

With long roots of squared length two, let I_λ be the trace index, normalized by $I_{\mathrm{ad}} = 2h^\vee$, and use the invariant inner product to define lattice duals. The audited formula is

$$D_\lambda^{\mathrm{wt}} := \sum_\mu \mathbb{Z}(m_\mu \mu) = Q_{\mathrm{rt}} + \mathbb{Z}\lambda, \qquad \boxed{\Lambda_\lambda = I_\lambda (Q_{\mathrm{rt}} + \mathbb{Z}\lambda)^*.}$$

Multiplicity weighting means multiplying a weight by m_μ, not merely repeating it in a list. This reduces the irreducible source calculation to root-lattice, central-character and index data, without enumerating a large representation. The proof uses multiplicity-one weights along suitable lowering paths. See the classification and its audit (SRC-02).

For G_2, F_4 it specializes to $I_\lambda Q_{\mathrm{long}}$. For E_6, E_7 it gives $I_\lambda X$ if $\lambda \in Q_{\mathrm{rt}}$, and $I_\lambda Q_{\mathrm{rt}}$ otherwise. For E_8, $X = Q_{\mathrm{rt}}$. Here Q_{long} is the lattice generated by the long roots, in this inner-product convention.

2.3 Comparisons and limitations already established

The coprime lattice/scaling lemma gives all-degree, all-Levi source and image comparisons for the specified E_6/E_7 minimal/adjoint pairs. The F_4 pair has the recorded degree-d scaling 3^d. Repeating the same representation leaves normalized trace projection unchanged. Arbitrary direct sums are subtler: three copies of the F_4 minimal module and the adjoint both have index 18, but their linear sources are $6Q_{\mathrm{long}}$ and $18Q_{\mathrm{long}}$, respectively. An exact combined-coefficient algorithm exists; a closed summand-level formula remains a supporting question (SRC-04/05/07, OPEN-04/05).

3 Completed image work and the link to Parts II and III

3.1 Classical generator formulas and integral refinements

In the notation of the papers, type C uses $\xi_{n,k}$, type B uses $\bar{\xi}_{n,k}$, and type D uses $\xi^D_{n,k}$. The targets are $\mathrm{IG}(n-k,2n)$, $\mathrm{OG}(n-k,2n+1)$, and $\mathrm{OG}(n-k,2n)$, respectively. The published generator and stable-injectivity results are inputs [3, 4]; the repository's combinatorial note gives alternative arguments and completes the dissertation proof route in its stated scope. It does not repair a missing proof of the published main theorems.

Our Smith results use the specified normalized tail generators

$$q_i = e_i(x_{n-k+1}^2, \dots, x_n^2), \qquad p = x_{n-k+1}\cdots x_n, \quad p^2 = q_k \quad \text{in type D.}$$

They address the arithmetic of their selected image, which is additional information beyond the complex generator formulas.

- **Type B:** a closed positive partition/Gaussian formula for every nonzero Smith multiplicity, in all allowed ranks and blocks, with the required integral primitive-prefix argument (IMG-03).
- **Type D:** the square-sector factors agree with type B; the full square/Euler formula includes the even-tail gluing step. The audited proof places gluing at unit levels, with no further positive-level correction (IMG-04/05).
- **Two-step flags:** normalized type C joint images are saturated and the separate-image sum is primitive. In B/D the separate sum is primitive inside the joint image, and the ambient factors are powers of 2; ambient saturation is false in general (IMG-08/09).

Here the B single-tail range has $a = n-k \geq 1,\ k \geq 1$. For D use $n \geq 4,\ a \geq 1,\ k \geq 2$, matching Part III. The literal defining/vector character forms and the normalized polynomial forms are distinguished explicitly; in the specified SO/Sp conventions the degree-d rescaling is 2^d, not an implicit change of lattice.

3.2 Exceptional images and independent checks

The completed finite cases include selected integral-image tables for all G_2/F_4 maximal parabolics, and the $E_6/P_1, E_7/P_7$ cases. The latter have primitive-generator subalgebras

$$\mathbb{Z}[h, 16s] \quad \text{and} \quad \mathbb{Z}[h, 81s, 19683t]$$

inside their known integral cohomology rings, with inherited relations. Here h is the Schubert divisor; s has polynomial degree 4 in E_6/P_1, while s, t have polynomial degrees $5, 9$ in E_7/P_7. They are not free polynomial rings and need not be saturated. Literal-character images and embedded minimal/adjoint comparisons are also computed for these two spaces. Minimal and adjoint images can already be incomparable in divisor degree.

Separately regenerated minimal-representation Springer coordinates agree with the printed F_4/E_7 formulas. The E_6 comparison proposes replacing the six denominators 6 by 18 in Part III, Lemma 7.1, with numerators unchanged. This is a documented proposed correction, not a published erratum. The original articles are preserved.

The research ledger, entries IMG-01–09, CASE-01–05 and LIMIT-01–05, links the exact proofs, source choices, tables and counterexamples. None of these finite tables is being used as an arbitrary-rank proof.

4 Completed geometry and the connection to Kostant–Michor

The normalized trace construction is the generalized Cayley map considered by Kostant–Michor [5]. Their quotient-degree comparison and harmonic localization theorem supply the geometric motivation. The same θ_ρ is being studied before passing to the Borel quotient.

4.1 Uniform formulas already available

In logarithmic torus coordinates, exact differentiation gives

$$J_T = M_\rho^{-1} \sum_\mu m_\mu \mu \mu^t e^\mu.$$

Equivariance determines the root-direction factors of the full, left-translated differential:

$$\boxed{\Psi_\rho|_T = \det(J_T) \prod_{\alpha>0} e^\alpha \left(\frac{\alpha(\theta_\rho)}{e^\alpha - 1}\right)^2.}$$

The root quotients extend as Laurent polynomials. Thus the Cartan Jacobian alone is not the full determinant. A tensor-Casimir identity also provides the quadratic invariant pullback; this applies standard representation theory rather than asserting a new Casimir identity (GEO-02–04).

4.2 Exact exceptional calculations already completed

Complete basic-invariant pullbacks and full determinant products are packaged for G_2, E_6, E_7, minimal and adjoint representations. The E_6/E_7 coordinates were checked against independent representation matrices and exact support/unisolvence certificates. Generic degrees on simply connected groups are recorded as follows:

	min	ad
G_2	6	18
E_6	1080	2754
E_7	7272	33664

The E_6/E_7 degree arguments use normalized weight-polytope volumes, boundary zero schemes and intersection contributions, restoring central cover factors where appropriate. Newton volume without correction gives the wrong answer in three of these four cases.

Seven specialized finite-field fiber reports corroborate the E_6/E_7 calculations: two primes for each of the first three cases and one for E_7 adjoint. The completed adjoint E_7 root-chart fiber has length 16832 at prime 32003; the central factor gives 33664. Its staircase length was recounted independently. These reports check specialized calculations; they are not the characteristic-zero degree proof. See the degree derivation and coverage and the E_6/E_7 geometry notes (CASE-06–08).

4.3 What these computations do not settle

The source critical divisor is computed; its image is not automatically a target branch-locus calculation, and nonproperness must be treated separately. The degree of a particular Springer morphism is not the minimum Cayley degree of the group. The two G_2 examples have regular-to-nonregular witnesses giving the recorded obstruction to removing harmonic localization. Comparable general criteria remain a goal. These distinctions define Aims B and C.

5 Aim A: a local theory of integral Springer–Borel images

Status: Primary project. General source classification and several image families are available; a uniform image criterion is not.

Question. For the literal source $\mathcal{A}_{\rho,L}$, or a separately declared normalized generator algebra, predict the prime support of D_d, give saturation criteria, and then determine selected prime-power multiplicities. The source choice is part of the problem, not an interchangeable convention.

A1. Establish a local benchmark and attribute its known content. Put $K = X \cap \Lambda_\rho$ and $\delta_\rho = [X : K][\Lambda_\rho : K]$. At a prime $p \nmid \delta_\rho$, the two linear lattices agree after localization at p. The first target is a precise comparison of their Levi invariant sources and characteristic images. Determine which additional hypotheses, involving the characteristic map and Levi descent, imply

$$\beta^P\Big(\mathrm{Sym}_{\mathbb{Z}_{(p)}}(\Lambda_\rho \otimes \mathbb{Z}_{(p)})^{W_L}\Big) = H^{2*}(G/P, \mathbb{Z}_{(p)}).$$

This is a *target statement with hypotheses to be supplied*, not a theorem asserted here. Classical torsion-index results and averaging at primes not dividing $|W_L|$ give a conservative proof route [6]. First identify the part that is already a standard corollary. The research question is how far one can remove conservative exclusions, and what happens at the remaining primes.

A2. Separate three possible arithmetic obstructions. Compare, in fixed coordinates, the change $X \rightsquigarrow \Lambda_\rho$, the integral invariant-ring calculation for W_L, and the Borel kernel and integral Schubert lattice. Localize before taking a Smith form. Use explicit invariant presentations or compatible bases to show which obstructions survive in the image; do not assume their valuations simply add. Work degreewise initially, and seek a ring-level criterion when possible.

A3. Resolve the first genuinely difficult shared-prime comparison. The coprime comparison lemma is already proved. Choose an exceptional representation pair with a shared-prime obstruction and derive image containment, sums or intersections over $\mathbb{Z}_{(p)}$, accounting for the Borel kernel. Begin on E_6/P_1 or E_7/P_7, where ambient rings and independent calculations are available. Only then test additional parabolics that distinguish competing mechanisms.

Proof tools and deliverables. Use the highest-weight lattice formula, integral invariant theory, characteristic-map/torsion-index results, Schubert evaluation, and exact lattice comparisons. Deliver (i) a proved local comparison with exact scope, (ii) a finite set of possible obstruction primes under stated hypotheses, and (iii) one sharp exceptional example with an explanation of its exponents. A counterexample to a stronger saturation statement is a valid alternative to (iii), provided its mechanism is proved.

Decision gate. If the first prime bound follows entirely from existing results, record it as such and proceed to shared-prime exponents or remove an unnecessary prime exclusion. Do not present a standard good-prime corollary as the main new result. Related ledger questions: OPEN-02/04/05/09/10/13.

6 Aim B: effective generic-degree formulas from boundary geometry

Status: The general intersection framework is available, and selected exceptional cases are worked out; an effective uniform evaluation is missing.

Question. Compute $\deg \theta_\rho$ from weight and root data with an explicit account of face degeneracies and their multiplicities. Seek a family theorem or a certified algorithm whose output is a proof certificate, not a guessed volume.

Write $\chi_\rho = \sum m_\mu e^\mu$ and $D\chi_\rho = \sum m_\mu \mu e^\mu$. The invertible trace matrix does not affect a generic degree. On the appropriate character lattice M, the equations are $D\chi_\rho = b$ for generic b. Kostant–Michor identifies the full degree with the adjoint-quotient degree; the two Weyl quotients have the same order, so no extra division by $|W|$ is made [5].

B1. Turn the known reduction into an effective procedure. For $\Delta = \mathrm{conv}(\mathrm{supp}\, D\chi_\rho)$, toric intersection theory supplies total degree $r!\,\mathrm{vol}_M(\Delta)$ [8]. The task is to compute all boundary contributions. On an exposed proper face F, study the scheme cut out by the face character and its logarithmic derivatives:

$$\chi_F = 0, \qquad D\chi_F = 0.$$

Account for the face lattice and finite covers, nonreduced structure, positive-dimensional loci, and overlaps of strata. A symbolic *volume minus boundary* identity is a starting framework; the desired result makes its correction terms independently evaluable.

B2. Prove the mechanism for a coherent family. Begin with minuscule representations. The E_6/E_7 examples already distinguish vanishing and nonvanishing face contributions, so minuscule weights do not automatically mean no boundary correction. Compare their faces with type A exterior-power weight polytopes. Classify the singular face schemes in a restricted family before claiming a formula for every minuscule representation. Adjoint root polytopes give a second family once the first mechanism is understood.

B3. Validate the degree without conflating evidence and proof. Check lattice volumes by determinant triangulation, face orbits by exact root combinatorics, and local multiplicities by an independent algebraic description where feasible. Use exact specialized fibers, retaining nilpotents, as corroboration. Specialization requires separate justification if it is to enter a characteristic-zero proof. Restore a central-isogeny degree only when the map actually factors through that cover.

Deliverable and fallback. The preferred result is a finite face-scheme/intersection procedure with proved correctness for the selected family, plus a closed formula when its strata admit one. A useful fallback is a complete theorem for simplex and cross-polytope face patterns under explicit smoothness and intersection hypotheses. F_4, additional minuscule cases or E_8 are tests selected for their face geometry; an expensive E_8 computation is not a milestone by itself. Related questions: OPEN-01/11/13/16.

7 Aim C: regularity, critical divisors and harmonic separation

Status: A universal determinant product and two G_2 obstruction witnesses are available. No general representation criterion is claimed.

Question and consequence. A regular element has centralizer dimension $\operatorname{rank} G$. When does θ_ρ send every regular group element to a regular Lie-algebra element? Kostant–Michor's Theorem 3.2 identifies this condition with their unlocalized harmonic-separation identity [5]. The target is a classification for a family, or a reusable obstruction, rather than a repeated determinant expansion.

Route. For regular semisimple $t \in T$, failure is witnessed by $e^{\beta}(t) \neq 1$ for every root β, but $\alpha(\theta_\rho(t)) = 0$ for some root α. Analyze these Laurent equations and seek a root-subsystem construction that lifts from small rank. Any lifting must prove that unwanted root directions remain regular and that the full representation restricts as required; a small-rank example alone does not supply this.

A semisimple witness disproves preservation. A positive theorem must also treat regular elements with nontrivial unipotent part, using Jordan decomposition and the published structure theory. Neither nonvanishing of the Cartan Jacobian nor a list of full critical factors is equivalent to that classification.

Deliverables. First recover the G_2 obstruction through a formulation usable in other types. Then prove a sufficient condition or an obstruction for a specified representation family, using $F_4/E_6/E_7$ as tests. Target branch and nonproperness loci are subsequent projects, not prerequisites. Related questions: OPEN-06/11/12.

8 Aim D: an integral geometric explanation of Smith filtrations

Status: Supporting project, activated when it helps explain Aim A. The B incidence/defect interpretation is a concrete starting point.

Hiller's chain viewpoint relates Schubert combinatorics to intersections [7]. The proposed bridge is more demanding than matching Betti numbers: one needs a map to the *integral coefficient-row filtration* of the Springer image, with primitive prefixes and the correct powers of 2.

Choose one chain/incidence model and one B single-tail family. Construct an integral comparison to the existing Schur-product prefixes, prove primitivity geometrically, and recover the known Smith formula as a test. Only after that attempt D's Euler gluing or B/D two-step multiplicities. The completed even-tail theorem is not being reopened; the goal is a conceptual explanation or a transferable proof. If only rational ranks match, record that as insufficient, and keep the current proof. Related questions: OPEN-03/07/08.

9 Track E: other cohomology theories and groups

Status: Newly adopted research directions. The ambient theories cited below are established; representation-dependent image extensions are proposed. No new backend or general extension theorem is claimed.

E1. Equivariant integral images: first extension to activate

Retain the same finite-dimensional group and Springer source, but replace ordinary cohomology by $H_T^*(G/P, \mathbb{Z})$. Put $S = H_T^*(\mathrm{pt}, \mathbb{Z}) = \mathrm{Sym}_{\mathbb{Z}}(X)$. The rational equivariant characteristic map $\beta_T^P : \mathrm{Sym}_{\mathbb{Q}}(V)^{W_L} \to H_T^{2*}(G/P, \mathbb{Q})$ gives a candidate S-subalgebra generated by $\beta_T^P(\mathcal{A}_{\rho,L})$, provided integrality is justified. The equivariant parameters and polynomial Cartan inputs are different copies of the weight variables.

The objective is to describe this embedded image as an S-module or algebra, and explain its specialization to the ordinary image. Positive-degree global invariant polynomials no longer simply vanish: in the rational equivariant Borel presentation, their Cartan values agree with the corresponding equivariant-parameter values. Fixed-point restrictions and generalized Demazure operators provide established computational methods [9].

Route and first test. Start with one small complete flag, then a parabolic already understood integrally. Construct equivariant generator restrictions and a certified module presentation. Compare exact fixed-point data with independent equivariant Schubert multiplication. Check both the ordinary specialization and the source normalization. Use G_2 or a low-rank classical example to distinguish mechanisms before returning to large exceptional groups.

Arithmetic and the specialization problem. The polynomial ring S is not generally a PID, so there is no ordinary Smith classification over S. Investigate Fitting ideals, local presentations and finite total-degree $\mathbb{Z}$-lattices. Distinguish torsion as an S-module from torsion over $\mathbb{Z}$. The natural specialization of the image maps onto the ordinary image when the chosen source generators specialize as intended; it need not identify the specialized image module injectively. Kernel/base-change phenomena must be recorded rather than inferred from a matching table.

Milestone and gate. Deliver a precisely defined integral equivariant image in one family, its module presentation, and a proved specialization comparison. A specialization defect with an exact explanation is an acceptable result. Activate after Aim A's local diagram has fixed the source and integrality conventions; do not wait for every ordinary Smith factor to be classified. Proposed API objects should preserve the coefficient ring, parameters, presentation and specialization maps (OPEN-14).

Longer objective. Determine whether equivariant data explains the ordinary prime support or filtration. A claim of equivariant saturation must specify the relevant module property; it is not an automatic translation of an ordinary integer-lattice statement.

E2. K-theory: compare character images through filtrations

The associated-bundle map

$$\kappa^P : R(L) \longrightarrow K^0(G/P), \qquad [U] \longmapsto [G \times_P U]$$

is defined by extending an L-representation trivially across the unipotent radical of P. Restrict it to the literal integral Springer polynomial-character subring

$$R_{\rho,\mathrm{poly}}(L) = \theta_\rho^*(\mathcal{A}_{\rho,L}) \subset R(L).$$

This produces an integral image to investigate without postulating a new Springer morphism. Its equivariant version takes values in $K_T^0(G/P)$, over the character ring $R(T) = \mathbb{Z}[X]$. Existing equivariant K-Schubert calculus supplies the ambient operators and classes [10].

Question and route. Do the cohomological divisibility factors occur as associated-graded features of this K-image? Compare the induced augmentation/gamma or geometric filtrations, with the rational Chern character as a diagnostic. The Springer polynomial grading is not automatically the augmentation filtration, and the rational Chern character does not prove an integral identification. State the filtration, prove its compatibility, and measure any discrepancy.

Milestone. For one family already covered by the ordinary API, construct the restricted K-image and compute the filtered comparison. An exact obstruction to identifying its associated graded with $J_{\rho,L}$ is as valuable as a positive theorem. Proposed library support comprises associated-bundle evaluation, K-Demazure operations, filtered image presentations and comparison reports. This is OPEN-17, and can start after the source conventions in Aim A without depending on a complete equivariant implementation.

E3. Oriented theories: a formal-group-law framework

An oriented theory uses

$$c_1(\mathcal{L} \otimes \mathcal{M}) = F(c_1(\mathcal{L}), c_1(\mathcal{M})).$$

Ordinary cohomology has $F(u, v) = u + v$; connective K-theory has $F(u, v) = u + v - \beta uv$. Algebraic cobordism supplies a universal formal group law. Formal group algebras and formal Demazure operators already organize characteristic maps for these theories [6, 9].

Main problem and first target. Define a representation-dependent source inside this framework and compare its images under coefficient specialization. Our degree-one theorem concerns the additive symmetric algebra; it is not already a theorem for every formal group law. A rational logarithm may guide the comparison but introduces denominators and does not settle integral sources. Start with connective K-theory, explicitly specifying the coefficient ring, completion and group-form hypotheses. Prove either a compatible source/image specialization to the additive case or an obstruction to such compatibility. Extend to algebraic cobordism only after that pilot succeeds. This is the unifying longer goal behind E1/E2 (OPEN-18).

E4. Quantum images: deform the map and its kernel

The proposed target is a lift $\xi^P_{\rho,q}$ with specialization at $q = 0$ equal to our ordinary map. This notation denotes a goal, not a currently defined general homomorphism. Specify its source, coefficient ring and generator lifts before asking for its integral image. Ordinary relations cannot be retained unchanged; for example,

$$H^*(\mathbb{P}^1, \mathbb{Z}) = \mathbb{Z}[h]/(h^2), \qquad QH^*(\mathbb{P}^1) = \mathbb{Z}[h, q]/(h^2 - q).$$

A map from an appropriate polynomial source can exist while its classical kernel changes. A ring section from ordinary to quantum cohomology is not provided by identifying their Schubert bases.

Route and benchmark. Begin with complete flags and the published quantum Borel/Toda presentations [11]. Fix divisor conventions and restrict a certified polynomial-source evaluation using quantum products. Then address a partial-flag family, where higher-degree generator lifts need compatible relations. Use quantum Giambelli/Pieri or Chevalley formulas as established ambient input, not as proof that the Springer-image construction has lifted. Track the integral normalization inherited from the original source.

Milestone and gate. Construct one multiplicative lift with a proved $q = 0$ specialization, then compare its kernel and image with the classical ones. Study flatness or base-change defects rather than assuming them away. Total-degree pieces can support finite integer-lattice checks when the chosen grading and coefficient ring make those pieces finite free. A failed candidate lift should yield an exact relation obstruction. E4 starts after a source and presentation are fixed, not after merely implementing quantum products (OPEN-15).

E5. Affine Grassmannians: transport image questions

The Peterson correspondence, proved by Lam–Shimozono, relates $QH^*_T(G/P)$ to a quotient of equivariant affine-Grassmannian homology after specified localizations [12]. Its product is the Pontryagin homology product. This is not an identification with ordinary cup-product cohomology of the affine Grassmannian.

Route and milestone. Once E4 supplies an image, transport it across the established correspondence for a chosen G/P. Record the exact quotient, inverted classes and coefficient lattice. Describe the transported generators, then determine what image information is lost under localization. An integral image statement before localization requires additional justification; a localized correspondence alone does not recover it. The first deliverable is a commutative comparison and an exact localized image description for one family (OPEN-19).

Separately, Mare–Mihalcea construct particular affine quantum deformations related to periodic Toda systems [13]. These are relevant test frameworks, not a claim that every desired quantum theory of an arbitrary infinite-dimensional flag variety has already been defined.

E6. Kac–Moody flag varieties and extensions of the group

There are two distinct projects. The BGG/nil-Hecke characteristic-map side already has Kac–Moody foundations through Kostant–Kumar [14]. A degree-bounded backend can use nil-Hecke operators, finite Schubert varieties or specified graded completions, with the root datum and coefficient lattice retained. The finite-type coinvariant presentation must not be imposed unchanged on an infinite Weyl group.

The Springer side needs a new construction. An infinite-dimensional Kac–Moody Lie algebra cannot inject into the endomorphisms of a finite-dimensional vector space. Highest-weight modules are typically infinite-dimensional, so ordinary finite traces and the finite matrix $M_\rho = \sum m_\mu \mu\mu^t$ do not transfer directly. An invariant bilinear form on a symmetrizable algebra by itself does not define the desired group-to-Lie-algebra trace projection.

Route and first deliverable. First distinguish affine loop groups from general symmetrizable Kac–Moody groups and choose a precise Cartan realization, weight lattice, topology/completion and class of representations. Develop the characteristic-map calculation independently of a new Springer map. If a completed or graded-character analogue of θ_ρ is proposed, prove that it is well defined, equivariant and compatible with a finite-type test or restriction. Do not call a formal infinite sum a regular morphism. The deliverable can be a rigorous domain and obstruction statement before any image theorem (OPEN-20).

Finite-dimensional group changes remain useful. The other-isogeny and reductive questions in OPEN-10 are closer extensions: replace $X(T)$ by the actual character lattice, handle central torus directions, and state trace-nondegeneracy hypotheses. Changing from SO to Spin is an arithmetic change of source, even when the underlying flag varieties agree. Positive-characteristic analogues would require separate existence, separability and trace-form analysis; they are not supplied by reducing our characteristic-zero formulas.

Activation order and success criteria. Start E1 after Aim A fixes its local source diagram. Use E2 as a filtered comparison pilot and E3 as the longer unifying direction. Define E4 carefully, then use E5 as the affine bridge. E6's nil-Hecke computations may proceed independently, but its Springer construction is a separate foundational project. At each stage, distinguish a known ambient theory from a new representation-image assertion. Each pilot must yield a proved comparison, an exact obstruction, or a precisely defined remaining gap. None of E1–E6 is marked as implemented or proved by this proposal.

10 Execution plan, milestones and computational support

The stages below are dependency gates, not elapsed-time promises. Start with Aim A; advance B as a distinct geometric project once its next proof obstruction is precise. C and D begin with bounded conceptual tests. Track E follows the activation order above, starting with equivariant images. Its later subprojects depend on their own source-definition and specialization gates, rather than launching together.

Stage	Work and output	Gate for moving on
0	Freeze notation/source forms; build theorem-to-source comparison against Demazure/CPZ, Kumar, Hiller and Kostant–Michor.	Each proposed claim has explicit hypotheses and a published/repository/open label.
1	Prove Aim A's conservative local benchmark; compute the candidate prime set from lattice and characteristic-map data.	Proof separates the three arithmetic stages and identifies standard content.
2	Resolve one shared-prime exceptional comparison; derive or disprove a sharper exponent/saturation prediction.	Independent embedded-lattice verification; theorem or exact counterexample.
3	Audit and generalize the face mechanism for Aim B in a chosen family.	All contributing strata and multiplicities accounted for; no raw-volume shortcut.
4	Run the regularity-obstruction and chain-filtration pilot arguments.	Reusable criterion/comparison, or a precise obstruction to the proposed route.
5	Package established statements, certificates and limitations.	Documentation and API expose proof scope; open questions remain separate.

First concrete work package. The next research session should (1) specify the local source/image diagram, (2) identify the exact characteristic-map and Levi-descent hypotheses, (3) prove the conservative comparison, and (4) use the existing exceptional images to test whether each excluded prime is necessary. The output should be a short proof note and an exact example matrix, not a new large table campaign.

Existing infrastructure. The `sage_borel` library already supplies Borel/BGG evaluation, classical coinvariant normal forms, arbitrary supported standard parabolics, rational kernels and finite-border certificates, Schubert multiplication, inverse Borel lifts, integral image matrices and Smith forms, membership and integer preimages, and embedded sums/intersections. Highest-weight source lattices and the packaged exceptional geometry connect these tools to the representation input (API-01–06).

Proposed API additions, conditional on the mathematics. Add a local source comparison report containing ρ, lattice bases, normalization, W_L, degree, coefficient ring and proved prime exclusions. Add a prime-specific image comparison returning embedded lattices and quotient invariants, and a boundary certificate listing face lattices, scheme equations and multiplicity justifications. A regularity witness should record the exact torus point, root evaluations and characteristic-zero verification. Return an explicit unresolved status when the requisite theorem is absent. These are proposed interfaces; this document does not claim they are implemented.

Resources. Ordinary lattice and low-degree checks use existing Sage routines. Exceptional elimination uses optional exact solvers with retained inputs, leading ideals and logs. No external computation budget is committed. Choose an alternative formulation or smaller discriminating case when an elimination becomes costly; a stalled run is not a result.

11 Risk control, scope and evidence standards

Mathematical risks and useful fallback results. Integral invariant rings can obscure a simple source classification. Fallback: a prime-local theorem or one Levi family. Boundary schemes may be nonisolated or overlap. Fallback: a family with proved transversality/local models, rather than dropping those strata. Regularity preservation may fail broadly. Fallback: a general obstruction with exact witnesses, rather than seeking a positive theorem at all costs. A chain model may recover ranks without integral primitivity. Fallback: identify the missing comparison explicitly and retain the existing proof.

Verification and literature. Every general theorem needs a written argument with its integral steps exposed. Verify decisive examples through a different realization: character/coefficient calculations versus representation matrices; coinvariant arithmetic versus independent cohomology presentations; geometric degree arguments versus exact fiber calculations. Agreement between implementations sharing the same input is a weaker check. Before a novelty claim, compare the exact statement and hypotheses with the relevant literature; an unsuccessful search does not establish originality.

The current baseline is repository commit `e9fcb7f` (20 October 2026), not an assertion about what has merged to the default branch. The runtime is version 0.20.0. The existing release record binds 157 regression checks and 24 isolated-wheel checks to runtime hashes; documentation verification records 40 executable examples and 908 local links/anchors. These establish computational coverage, not proof validity, external review or research priority.

Tracking progress. Use the research ledger and detailed proof-status record. For each milestone record: exact question; hypotheses; proof status; independent checks; source/novelty comparison; remaining obstruction; next discriminating test. Mark a theorem complete only when its general argument is supplied. A finite example stays a finite example. The proposal itself adds no mathematical theorem.

Ledger questions	Role in this program
OPEN-02/05/09	Core Aim A: arithmetic, shared primes and Levi images.
OPEN-04/10	Source extensions supporting A: direct sums and isogeny/reductive scope.
OPEN-01	Core Aim B: effective generic-degree calculation.
OPEN-06/12	Core Aim C: critical divisors and regularity/localization.
OPEN-03	Core Aim D: integral chain/incidence explanation.
OPEN-07/08	Conditional next applications: B/D joint factors and many Levi blocks.
OPEN-11/13	Target geometry and further exceptional tests, chosen by a mechanism.
OPEN-14/17/18	Track E1–E3: equivariant, filtered K-theoretic and formal-group-law images.
OPEN-15/19/20	Track E4–E6: quantum lifts, affine transport and Kac–Moody characteristic/Springer constructions.
OPEN-16	Deferred: comparisons with minimum Cayley degrees.

Expected contribution. The intended advance is a passage from explicit maps to a local arithmetic theory of their images, together with an effective geometric degree method. The completed source classification provides the representation input; the classical and exceptional examples provide demanding tests. A successful program explains which arithmetic and geometric features are forced, which depend on the representation or parabolic, and where a proposed general rule fails.

Sources and reproducible starting material

The published bibliography below supplies the construction and background. Repository notes are unpublished working documents, with internal audits. Their presence here does not establish priority. The original article PDFs and their provenance are indexed in the article collection. The detailed literature tasks at Stage 0 remain necessary; this is a source-grounded proposal, not an exhaustive literature review.

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[5] B. Kostant and P. W. Michor, *The generalized Cayley map from an algebraic group to its Lie algebra*, in *The Orbit Method in Geometry and Physics*, Progress in Mathematics **213** (2003), 259–296. arXiv:math/0109066; especially Theorem 3.2 and Corollary 3.3.

[6] B. Calmès, V. Petrov and K. Zainoulline, *Invariants, torsion indices and oriented cohomology of complete flags*, Ann. Sci. Éc. Norm. Supér. (4) **46** (2013), no. 3. arXiv:0905.1341. Includes the relation to Demazure's characteristic-map theory; all coefficient hypotheses must be retained when applying it.

[7] H. Hiller, *Combinatorics and intersections of Schubert varieties*, Comment. Math. Helv. **57** (1982), 41–59. journal archive. A chain/intersection background source; an integral Smith comparison is a proposed task, not a theorem attributed to this paper.

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Working-note map at the baseline commit.

- Source arguments: degree-one theorem, source audit, irreducible lattice formula, lattice audit.
- Classical arithmetic: B theorem, D square comparison, D full gluing theorem, C joint images, B/D joint images.
- Exceptional arithmetic: selected minuscule images, E6 character form, E7 character form.
- Geometry and computation: general/G2 geometry, E6 geometry, E7 geometry, degrees and verification coverage, release audit.